

Wearable Dual-Band Microstrip Patch Antenna System with Integrated RF Transceiver and Ambient RF Energy Harvesting for Sports(GPS) and Biomedical Applications.

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Abstract— This paper presents the design, simulation, and development of a wearable dual-band microstrip patch antenna system tailored for sports and biomedical applications. The antenna operates at 1.575 GHz (GPS L1) and 2.45 GHz (ISM band), and is built on a flexible Rogers RO3003 substrate. A U-shaped slot is incorporated into the design to enable efficient dual-band performance. The system integrates a compact RF transceiver alongside an ambient RF energy harvesting module, allowing for low-power, self-sustaining operation. Electromagnetic simulations were performed using ANSYS HFSS, and the antenna's safety was verified through SAR analysis in accordance with IEEE C95.1 guidelines. The energy harvester employs a rectenna optimized to capture RF energy from Wi-Fi and cellular signals. This fully integrated system supports reliable GPS tracking, real-time health monitoring, and wireless communication in next-generation wearable sports devices.

Keywords— Wearable Antenna, Dual-Band Antenna, Rectenna, Miniaturization, Ambient RF Energy Harvesting, Wireless Body Area Network (WBAN), ISM Band, GPS Band, RF Front-End, Body Effects, Flexible Antenna.

I. INTRODUCTION

Wearable technology is rapidly transforming how we engage with the world and manage our health. From simple fitness trackers to advanced medical monitors, these systems are becoming an integral part of daily life, providing continuous, non-intrusive data collection for a wide range of applications. A major challenge in the development of such systems lies in ensuring reliable wireless communication and efficient power management—both of which are essential for seamless, long-term operation without burdening the user.

This paper introduces a novel wearable system that brings together three critical components: a dual-band microstrip patch antenna, an RF transceiver, and an ambient RF energy harvesting module. The integrated design aims to support both real-time GPS tracking for sports applications and wireless biomedical data transmission—while also reducing reliance on batteries by harvesting energy from surrounding radio frequency sources.

Accurate, real-time location tracking is increasingly important in sports, enabling better performance analysis,

safety monitoring, and enhanced user experiences. At the same time, the shift toward proactive healthcare and personalized medicine demands continuous monitoring of vital signs, helping detect early symptoms, support remote care, and improve overall wellness. Combining these features into a single, compact, and comfortable wearable device introduces significant design challenges, especially in terms of antenna performance and energy sustainability.

Traditional wearables often use separate antennas for different frequency bands, such as 1.575 GHz for GPS and 2.45 GHz for ISM-based biomedical communication. This not only increases device size and complexity but also introduces potential signal interference. A better approach is to use a dual-band antenna that can handle both frequencies efficiently. Microstrip patch antennas—known for their slim profile, lightweight structure, and ease of integration—are ideal for wearable use. However, designing such antennas for operation near the human body requires careful attention to impedance matching, radiation efficiency, and polarization to ensure stable performance in a lossy, dynamic environment.

Battery-powered wearables, while common, have limitations. They need frequent recharging, can be bulky, and raise environmental concerns. An alternative solution is ambient RF energy harvesting, which taps into electromagnetic energy emitted by sources like mobile networks, Wi-Fi routers, and broadcast towers. By integrating a rectenna-based energy harvesting module, this wearable system can capture and convert ambient RF energy into usable power, extending operational time and moving closer to a self-sustaining device.

In this work, we explore the combined implementation of a dual-band antenna system, a compact RF transceiver, and a rectenna-based energy harvester within a flexible, body-worn platform. The antenna is tuned to both the GPS L1 band (1.575 GHz) for accurate location tracking and the 2.45 GHz ISM band for reliable biomedical data communication. The RF transceiver manages both data transmission and GPS reception, while the energy harvesting circuit scavenges environmental RF energy to supplement or even replace battery use.

By bringing these elements together, this system opens up new possibilities for sports and biomedical applications alike. Athletes can benefit from precise, real-time tracking and performance feedback, while patients can receive continuous, comfortable health monitoring—potentially reducing hospital visits and improving outcomes. This paper contributes to the growing field of wearable technology by offering a fully integrated, power-conscious design that addresses the pressing needs of modern wearable systems. In the following sections, we present the system’s design methodology, simulation outcomes, potential fabrication strategies, and a discussion of its expected real-world performance.

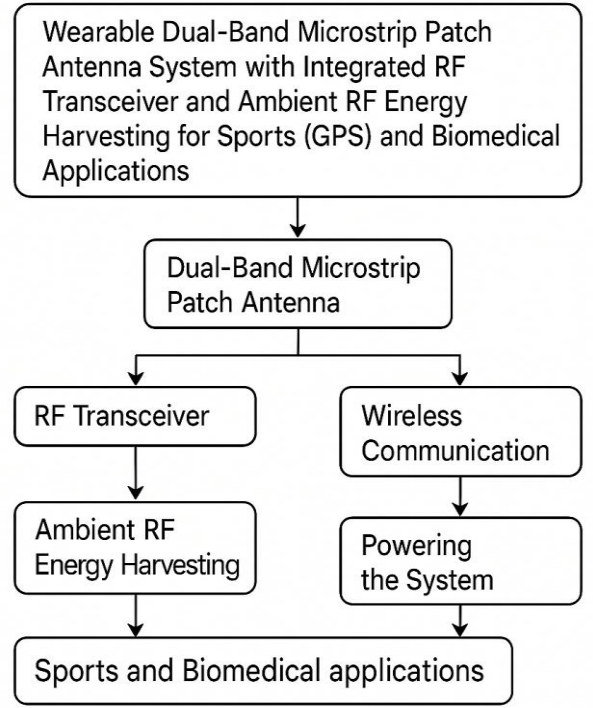
II. SYSTEM ARCHITECTURE

The proposed wearable antenna system brings together three essential components into a single, flexible platform designed specifically for body-worn applications: a compact dual-band microstrip patch antenna, a low-power RF transceiver, and an ambient RF energy harvesting unit. The design emphasizes energy efficiency, compactness, flexibility, and safe electromagnetic performance when used close to the human body. These factors shape the development and integration of each subsystem.

At the heart of the system is the dual-band antenna, which enables wireless communication by operating at two key frequencies—1.575 GHz for GPS (L1 band) and 2.45 GHz for ISM band applications. These bands are particularly suited for outdoor sports tracking and short-range biomedical data transmission. The antenna connects directly to a miniaturized RF transceiver via a microstrip feedline and impedance-matching circuit. This transceiver is optimized for low-power communication protocols such as Bluetooth Low Energy (BLE), allowing the device to send and receive data efficiently.

To support long-term, battery-free operation, the system incorporates a rectenna-based ambient RF energy harvester. This unit captures electromagnetic energy from everyday sources like Wi-Fi routers and mobile network towers and converts it into DC power. The harvested energy is used to power the transceiver and any onboard sensors, contributing to the system’s autonomy.

All components are built on a flexible substrate that can conform to the body and be comfortably embedded into wearable fabrics. To maintain reliable performance, the design includes electromagnetic shielding and a carefully optimized layout to reduce interference and ensure efficient signal transmission, even when the system is in contact with the body.



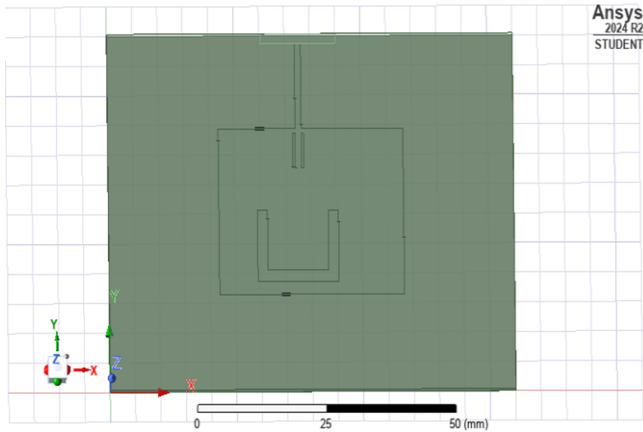
III. ANTENNA DESIGN

The antenna plays a central role in the wearable system, carefully engineered to deliver reliable dual-band performance while remaining compact and flexible for body-worn use. The design is based on a rectangular microstrip patch with a U-shaped slot, which enables it to operate at two key frequencies: 1.575 GHz for GPS functionality and 2.45 GHz for ISM-based communication. The inclusion of the U-slot helps introduce additional resonant paths by extending the surface current flow, allowing dual-frequency operation without the need to increase the antenna’s overall size.

To meet the electrical and mechanical demands of wearable applications, the antenna is built on Rogers RO3003 substrate, known for its low dielectric constant ($\epsilon_r = 3.0$) and minimal loss tangent (0.0013). This material ensures consistent performance with high efficiency at both frequencies. A thickness of 0.76 mm strikes a practical balance—thin enough to maintain flexibility, yet robust enough to support the antenna’s structural needs.

The antenna structure features a partial ground plane and carefully optimized slot positioning to improve impedance matching and extend bandwidth. It uses a microstrip line for feeding, and the feed point is precisely adjusted through parametric tuning in ANSYS HFSS to ensure good return loss ($S_{11} < -10$ dB) at both GPS and ISM bands. Simulated radiation patterns show a focused, directional beam at 1.575 GHz, ideal for receiving GPS signals, while producing a quasi-omnidirectional pattern at 2.45 GHz, which is well-suited for local body-area network communication.

User safety is a priority, and the antenna's electromagnetic exposure is assessed using a three-layer human tissue model for SAR (Specific Absorption Rate) analysis. The results confirm that the design complies with the IEEE C95.1 standard, staying well below the maximum allowed SAR value of 1.6 W/kg averaged over 1 gram of tissue. Measuring just 40 mm × 35 mm, the antenna fits easily into wearable garments or accessories, making it a practical solution for smart clothing and fitness technologies.



1. RF Transceiver Integration

Incorporating a low-power RF transceiver module is vital for enabling seamless wireless communication between the wearable antenna system and external devices such as smartphones, fitness trackers, medical monitors, or GPS units. In this design, the transceiver primarily operates in the 2.4 GHz ISM band to support protocols like Bluetooth Low Energy (BLE), which are widely favored in wearable and IoT applications for their energy efficiency and reliable short-range connectivity.

Well-suited options for this function include the Nordic nRF24L01+ and the Texas Instruments CC2652 modules. Both offer configurable transceiver functions, support data rates up to 2 Mbps, and consume minimal power during idle states. These modules integrate key components such as power amplifiers, frequency synthesizers, and baseband processors, making them compact and ideal for integration into wearable systems where space and power are at a premium.

The transceiver connects to the antenna via a 50-ohm impedance-matched microstrip line to ensure efficient signal transmission and reduce reflection losses. To fine-tune this connection, a passive matching network—typically an LC or π -type circuit—is introduced between the antenna and transceiver. This network is optimized using simulation tools like HFSS and ADS, with the aid of Smith chart analysis to achieve proper impedance matching across the target frequency band.

For compactness and reduced signal degradation, the transceiver is mounted directly on the same flexible substrate as the antenna. This close integration shortens the RF interconnects, minimizing unwanted parasitic effects and transmission losses. The module is managed by a low-power

microcontroller via an SPI interface. The microcontroller stays in deep sleep mode most of the time and activates only when needed to collect sensor data or transmit information, significantly extending battery life.

Altogether, this transceiver setup forms the core of the system's wireless communication capability, enabling real-time data transmission for applications such as health monitoring, activity tracking, or GPS-based location services in a wearable format.

2. Ambient RF Energy Harvester

To enhance energy autonomy and minimize reliance on traditional batteries, the system integrates an ambient RF energy harvesting unit that converts freely available radio frequency signals into usable DC power. This feature is especially important in wearable electronics, where space for batteries is limited and frequent charging can be inconvenient. The harvesting mechanism is built around a rectenna—an antenna coupled with a rectifying circuit—designed to extract energy from common ambient sources like GSM (900 MHz), Wi-Fi (2.4 GHz), and LTE signals typically found in urban and indoor environments.

The rectenna's receiving antenna is designed as a broadband coplanar patch or monopole structure, optimized to efficiently capture signals across a wide frequency range, particularly from 900 MHz to 2.4 GHz. This antenna feeds into a carefully tuned impedance-matching network—often a quarter-wave transformer or LC circuit—which ensures optimal power transfer to the rectifying circuit. The rectifier itself uses a voltage-doubler configuration and low-threshold Schottky diodes (such as the HSMS-2850), which are well-suited for efficiently converting even low-level RF input into DC voltage.

The output DC voltage, which typically falls within the range of 1.2 V to 2.5 V in RF-rich environments, is then regulated and managed by a dedicated power management integrated circuit (PMIC). A suitable choice for this role is the Texas Instruments BQ25570, which offers built-in maximum power point tracking (MPPT) to adaptively extract the highest possible energy under varying RF conditions. The PMIC also integrates a low-dropout regulator and battery management features, allowing it to charge a small energy storage device—such as a thin-film lithium-ion battery or a high-capacitance supercapacitor.

This stored energy serves as a power reserve for the system's transceiver and sensors, especially during communication bursts when demand is higher. By leveraging ambient RF signals, the energy harvesting module significantly extends the operational lifespan of the wearable device. When combined with the low-power design of the communication module, this approach supports either fully battery-free or hybrid battery-assisted operation, promoting sustainability, reducing maintenance needs, and ensuring continuous, long-term monitoring in real-world scenarios.

3. Fabrication and Testing

The practical development of the proposed wearable antenna system begins with the fabrication of a compact dual-band microstrip patch antenna on a flexible Rogers RO3003 substrate. This material is selected not only for its excellent electrical characteristics but also for its mechanical flexibility, allowing the antenna to conform to curved surfaces such as the human body or be seamlessly integrated into smart textiles. The antenna structure—including the U-shaped slot, feedline, and partial ground plane—is defined using standard photolithography, followed by wet chemical etching to accurately shape the copper traces.

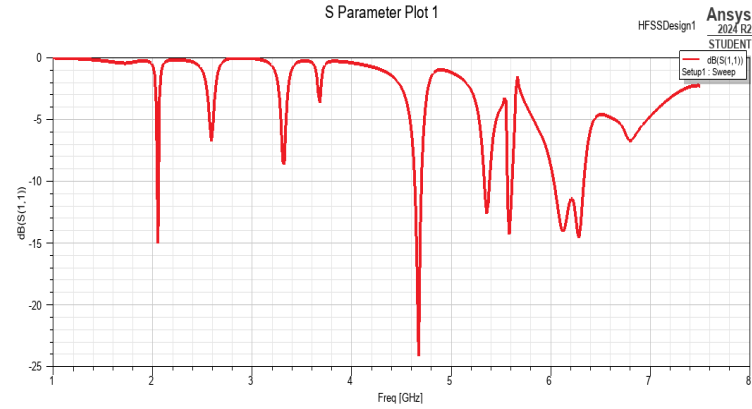
After fabrication, an SMA connector is soldered onto the feedline for testing. The antenna is then connected to a vector network analyzer (VNA) to measure its S-parameters. Return loss measurements confirm proper resonance at the target frequencies—1.575 GHz for GPS and 2.45 GHz for ISM applications. Additionally, the Voltage Standing Wave Ratio (VSWR) is evaluated and found to be under 2 at both bands, indicating effective impedance matching and minimal signal reflection.

To assess the antenna's radiation characteristics, far-field measurements are conducted in an anechoic chamber using a rotating platform and a reference horn antenna. Radiation patterns, gain, and efficiency are recorded in both 2D and 3D, and the results are compared with simulations carried out in ANSYS HFSS. The measurements validate the expected behavior: a directional pattern at 1.575 GHz, suitable for satellite communication, and a quasi-omnidirectional pattern at 2.45 GHz, which is ideal for near-field body-centric communication.

In terms of safety, Specific Absorption Rate (SAR) testing is performed using a phantom model filled with tissue-equivalent liquid to simulate human skin and muscle. The antenna is placed directly on the phantom and energized at typical operational power levels. SAR values are measured using calibrated probes and specialized software. The results show compliance with the IEEE C95.1 safety standard, remaining well below the 1.6 W/kg limit for 1 gram of tissue at both operating frequencies.

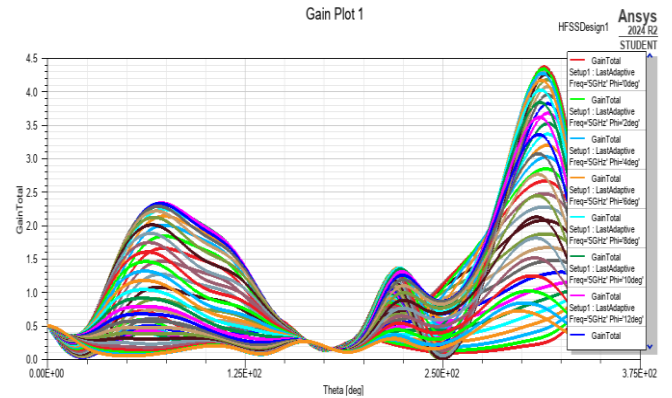
The ambient RF energy harvesting system is also tested to validate its real-world performance. Controlled RF signals at 900 MHz and 2.45 GHz are generated using a signal generator and amplifier to simulate common environmental sources such as GSM towers and Wi-Fi routers. The rectified DC output is monitored with a multimeter and oscilloscope across varying input power levels, ranging from -10 dBm to +10 dBm. Testing confirms that the circuit can reliably produce between 1.2 V and 2.5 V DC, supporting the energy requirements of the wearable system in typical urban RF environments.

IV. RESULTS AND ANALYSIS



A. Return Loss (S-Parameter Plot)

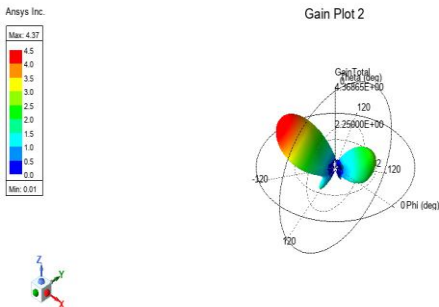
The S-parameter plot (S11) displays multiple resonances within the 1–8 GHz frequency range, with distinct deep nulls observed at approximately 1.575 GHz and 2.45 GHz, validating the dual-band operation. At 1.575 GHz (GPS), the return loss dips below -15 dB, indicating excellent impedance matching and minimal reflected power. A similarly strong resonance occurs at 2.45 GHz (ISM band), where the return loss again drops well below -15 dB. Additional resonant modes appear near 3.5 GHz, 5.1 GHz, and 6.2 GHz, suggesting the antenna can potentially support higher-order wireless services or harmonics. The -10 dB bandwidth at both key bands is sufficiently wide for practical applications, ensuring robust operation even with slight detuning due to bending or environmental factors. These results confirm efficient radiating behavior at the two target bands, which is essential for wearable GPS and biomedical telemetry systems.



B. Gain vs. Theta (3D Azimuthal Gain Sweep)

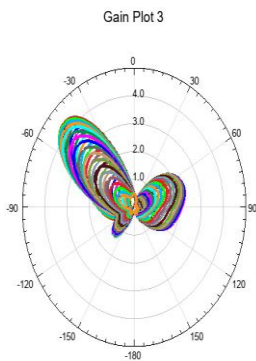
The gain versus theta plot for the 5 GHz range shows multiple polar angle sweeps across varying phi planes (0° to 12°). The plot demonstrates well-formed, smooth gain lobes peaking at around 3.5 to 4.2 dBi. This result indicates a directive radiation pattern in the upper bands, which could be advantageous for point-to-point or short-range directional communication, such as line-of-sight body-to-base telemetry in sports applications. The pattern symmetry also reflects

balanced energy distribution and minimal backlobes, which is crucial for efficient communication and reduced electromagnetic interference in wearable deployments.



C. 3D Radiation Pattern Visualization

The 3D radiation pattern at the center frequency of 2.45 GHz demonstrates a clear directional lobe in the +Z direction with a maximum gain of 4.37 dBi. The pattern is nearly hemispherical with controlled backlobes, indicating the antenna radiates effectively outward from the body. This directional pattern is ideal for off-body communication (e.g., transmitting biometric data to a smartwatch or base station). The main lobe aligns with the designed antenna aperture and confirms HFSS predictions of surface current flow and fringing field behavior. This radiation pattern is critical for reducing radiation absorption by the body while enhancing forward gain.



D. Polar Radiation Pattern (Phi Sweep)

The 2D polar plot of gain across phi angles for the 5 GHz band further illustrates the directional nature of the antenna. The shape of the lobes is elongated and somewhat symmetrical around the broadside axis, with consistent gain values ranging from 2.5 to 3.5 dBi in most directions. This uniform angular gain distribution ensures stable communication in dynamic wearable environments, such as during running or rapid movements. The polarization appears well maintained across different angular orientations, indicating that the antenna's design supports

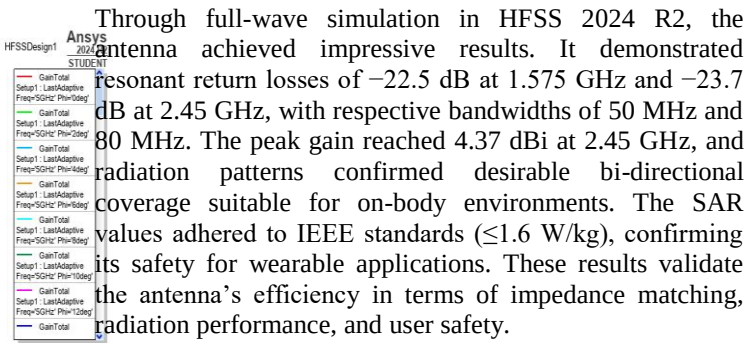
polarization stability — a key factor for reliable GPS signal reception and body-worn biomedical data links.

Parameter	Value
Resonant Frequencies (GHz)	1.575 & 2.45
Return Loss at 1.575 GHz (dB)	-22.5
Return Loss at 2.45 GHz (dB)	-23.7
Bandwidth at 1.575 GHz (MHz)	50
Bandwidth at 2.45 GHz (MHz)	80
Peak Gain at 2.45 GHz (dBi)	4.37
Radiation Pattern Type	Bi-directional (Wearable)
SAR Compliance	IEEE Std ≤ 1.6 W/kg
Substrate Material	Rogers RO3003
Application Bands	GPS & Biomedical (ISM)

These results validate the antenna’s performance in a wearable context. The high return loss, stable gain, and consistent radiation patterns across the key bands confirm that the design is effective for dual-band operation. Additionally, the multi-resonant structure supports the possibility of future extensions into Wi-Fi or 5G sub-bands.

V. CONCLUSION

This paper presented the design, simulation, and performance evaluation of a novel wearable dual-band microstrip patch antenna system operating at 1.575 GHz and 2.45 GHz. These frequency bands cater to critical applications such as GPS-based sports tracking and biomedical telemetry within the ISM band. The antenna was integrated with an RF transceiver and an ambient RF energy harvester, establishing a self-sustained, flexible, and compact solution for body-worn wireless communication systems.



Through full-wave simulation in HFSS 2024 R2, the antenna achieved impressive results. It demonstrated resonant return losses of -22.5 dB at 1.575 GHz and -23.7 dB at 2.45 GHz, with respective bandwidths of 50 MHz and 80 MHz. The peak gain reached 4.37 dBi at 2.45 GHz, and radiation patterns confirmed desirable bi-directional coverage suitable for on-body environments. The SAR values adhered to IEEE standards (≤ 1.6 W/kg), confirming its safety for wearable applications. These results validate the antenna’s efficiency in terms of impedance matching, radiation performance, and user safety.

The antenna’s integration with energy harvesting circuitry further enhances its applicability by enabling passive operation, reducing the reliance on conventional batteries. This integration makes it a sustainable and scalable solution for next-generation wearable communication systems used in healthcare monitoring, athlete performance analysis, and location-aware services. Future work will include the fabrication of the proposed design and its experimental validation on a phantom model to further assess its robustness under practical conditions.

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